ahb_sramc_seqlib.sv + (~/keller_work/test_ing/SRAMC_UVMTB/ahb_sramc_uvmtb/verif/tests/seqlib) - GVIM1

File Edit Tools Syntax Buffers Window UVM Verilog Help

```
22                                  req.op    == WRITE;} )
23        `uvm_error(get_type_name(), "Failed to randomize transaction")
24
25      $cast(temp_tr, req.clone());
26      temp_tr.op = READ;
27      rd_req.push_back(temp_tr);
28      finish_item(req);
29    end
30
31    foreach(rd_req[i]) begin
32      temp_tr = rd_req[i];
33      start_item(temp_tr);
34      finish_item(temp_tr);
35    end
36    req = ahb_seq_item#(32,32)::type_id::create("req");
37    start_item(req);
38    if ( !req.randomize() with {req.trans == IDLE;} )
39      `uvm_error(get_type_name(), "Failed to randomize transaction")
40    finish_item(req);
41
42    `uvm_info(get_full_name(), "Default seq completed.", UVM_HIGH)
43  endtask
44
45
46  class ahb_sramc_full_addr_space_AA_55_data_seq extends ahb_base_seq;
47
48    `uvm_object_utils(ahb_sramc_full_addr_space_AA_55_data_seq)
49
50    function new(string name = "ahb_sramc_full_addr_space_AA_55_data_seq");
51      super.new(name);
52    endfunction
```

40,20 34%

switch success

or] there

07:43 / 47:36 护眼: 关 x2倍速 高清

ahb_sramc_uvmtb_demo_test01.log (~/keller_work/test_ing/SRAMC_UVMTB/ahb_sramc_uvmtb/verif/sim) - GVIM8

File Edit Tools Syntax Buffers Window UVM Verilog Help

```
165383    [0]        ahb_rsp_type_e          2     OKAY
165384  beats_addr   da(integral)            0     -
165385  hxdata       da(integral)            0     -
165386  beat_hxdata  integral                32    'h95bf26de
165387  busy_cycle   da(integral)            0     -
165388 ----------------------------------------------------
165389
165390 UVM_INFO ../env/ahb_sramc_uvmtb_scb.sv(491) @ 26415000: uvm_test_top.m_env.m_scb [uvm_test_top.m_env.m_scb] SCB Report: match_counter=5,
       mismatch_counter=1595.
165391 ================================
165392 *********Testcase FAIL**********
165393 ================================
165394
165395 --- UVM Report Summary ---
165396
165397 ** Report counts by severity
165398 UVM_INFO : 7529
165399 UVM_WARNING :    0
165400 UVM_ERROR :    0
165401 UVM_FATAL :    0
165402 ** Report counts by id
165403 [RNTST]     1
165404 [UVMTOP]     1
165405 [ahb_monitor]     1
165406 [ahb_sramc_uvmtb_demo_test01]     2
165407 [ahb_sramc_uvmtb_env]     2
165408 [uvm_test_top]     1
165409 [uvm_test_top.m_env.m_ahb_agent.m_driver]  3685
165410 [uvm_test_top.m_env.m_ahb_agent.m_monitor]  3833
165411 [uvm_test_top.m_env.m_ahb_agent.m_sequencer.m_seq]     2
165412 [uvm_test_top.m_env.m_scb]     1
search hit BOTTOM, continuing at TOP
-- VISUAL --                                                    165390,1      99%
```

switch success

verwcj@VM01:~/ teaching_sim | ahb_seq.sv (~/kel... | ahb_sramc_uvmt... | ahb_sramc_seqli... | ahb_sramc_full_a... | ahb_driver.sv (~/... | ahb_sramc_uvmt... | ahb_sramc_full_a... | ahb_sramc_uvmt...

08:57 / 47:36

ahb_slave_if.v (~/keller_work/test_ing/SRAMC_UVMTB/ahb_sramc_uvmtb/rtl/ahb_sramc) - GVIM5

File Edit Tools Syntax Buffers Window UVM Verilog Help

```
150
151   //Generate sram write and read enable signals.
152   assign sram_write = ((htrans_r == NONSEQ) || (htrans_r == SEQ)) && hwrite_r;
153   assign sram_read =  ((htrans_r == NONSEQ) || (htrans_r == SEQ)) && (!hwrite_r);
154   assign sram_w_en = !sram_write;
155
156   //generate sram address
157   assign sram_addr = haddr_r [15:0];
158   assign sram_addr_out = sram_addr[14:2];
159
160   //Generate bank select signals by the value of sram_addr[15].
161   //Each bank(32kx32) comprises of four sram block(8kx8), and
162   //the width of the address of the bank is 15 bits(14~0), so
163   //the sram_addr[15] is the minimun of the next bank. If its
164   //value is "1", it means the next bank is selected.
165   assign sram_csn_en = (sram_write || sram_read);
166   assign bank_sel = (sram_csn_en && (sram_addr[15] == 1'b0)) ? 1'b1 : 1'b0;
167
168   assign bank0_csn = (sram_csn_en && (sram_addr[15] == 1'b0)) ? sram_csn : 4'b1111;
169   assign bank1_csn = (sram_csn_en && (sram_addr[15] == 1'b1)) ? sram_csn : 4'b1111;
170
171   //Choose the right data output of the two banks(bank0, bank1) according
172   //to the value of bank_sel. If bank_sel = 1'b1, bank0 sleceted, or
173   //bank1 selected.
174   assign  sram_data_out = (bank_sel) ?  {sram_q3, sram_q2, sram_q1, sram_q0} :
175                                         {sram_q7, sram_q6, sram_q5, sram_q4} ;
176
177   //signals used to generating sram chip select signal in one bank.
178   assign haddr_sel = sram_addr[1:0];
179   assign hsize_sel = hsize_r [1:0];
180
```

"../../rtl/ahb_sramc/ahb_slave_if.v" 252L, 7806C written

switch success

-- VISUAL --

174,74 67%

verwcj@VM01:... teaching_sim ahb_seq.sv (~/... ahb_sramc_uv... ahb_sramc_se... ahb_sramc_full... ahb_driver.sv (... [ahb_sramc_uv... [ahb_sramc_uv... <Verdi:nTrace... ahb_slave_if.v (...

ahb_slave_if.v + (~/keller_work/test_ing/SRAMC_UVMTB/ahb_sramc_uvmtb/rtl/ahb_sramc) - GVIM5

File Edit Tools Syntax Buffers Window UVM Verilog Help

```
150
151   //Generate sram write and read enable signals.
152   assign sram_write = ((htrans_r == NONSEQ) || (htrans_r == SEQ)) && hwrite_r;
153   assign sram_read =  ((htrans_r == NONSEQ) || (htrans_r == SEQ)) && (!hwrite_r);
154   assign sram_w_en = !sram_write;
155
156   //generate sram address
157   assign sram_addr = haddr_r [15:0];
158   assign sram_addr_out = sram_addr[14:2];
159
160   //Generate bank select signals by the value of sram_addr[15].
161   //Each bank(32kx32) comprises of four sram block(8kx8), and
162   //the width of the address of the bank is 15 bits(14~0), so
163   //the sram_addr[15] is the minimun of the next bank. If its
164   //value is "1", it means the next bank is selected.
165   assign sram_csn_en = (sram_write || sram_read);
166   assign bank_sel = (sram_csn_en && (sram_addr[15] == 1'b1)) ? 1'b1 : 1'b0;
167
168   assign bank0_csn = (sram_csn_en && (sram_addr[15] == 1'b0)) ? sram_csn : 4'b1111;
169   assign bank1_csn = (sram_csn_en && (sram_addr[15] == 1'b1)) ? sram_csn : 4'b1111;
170
171   //Choose the right data output of the two banks(bank0, bank1) according
172   //to the value of bank_sel. If bank_sel = 1'b1, bank0 sleceted, or
173   //bank1 selected.
174   assign  sram_data_out = (~bank_sel) ?  {sram_q3, sram_q2, sram_q1, sram_q0} :
175                                         {sram_q7, sram_q6, sram_q5, sram_q4} ;
176
177   //signals used to generating sram chip select signal in one bank.
178   assign haddr_sel = sram_addr[1:0];
179   assign hsize_sel = hsize_r [1:0];
180
1 change; before #1  87 seconds ago
```

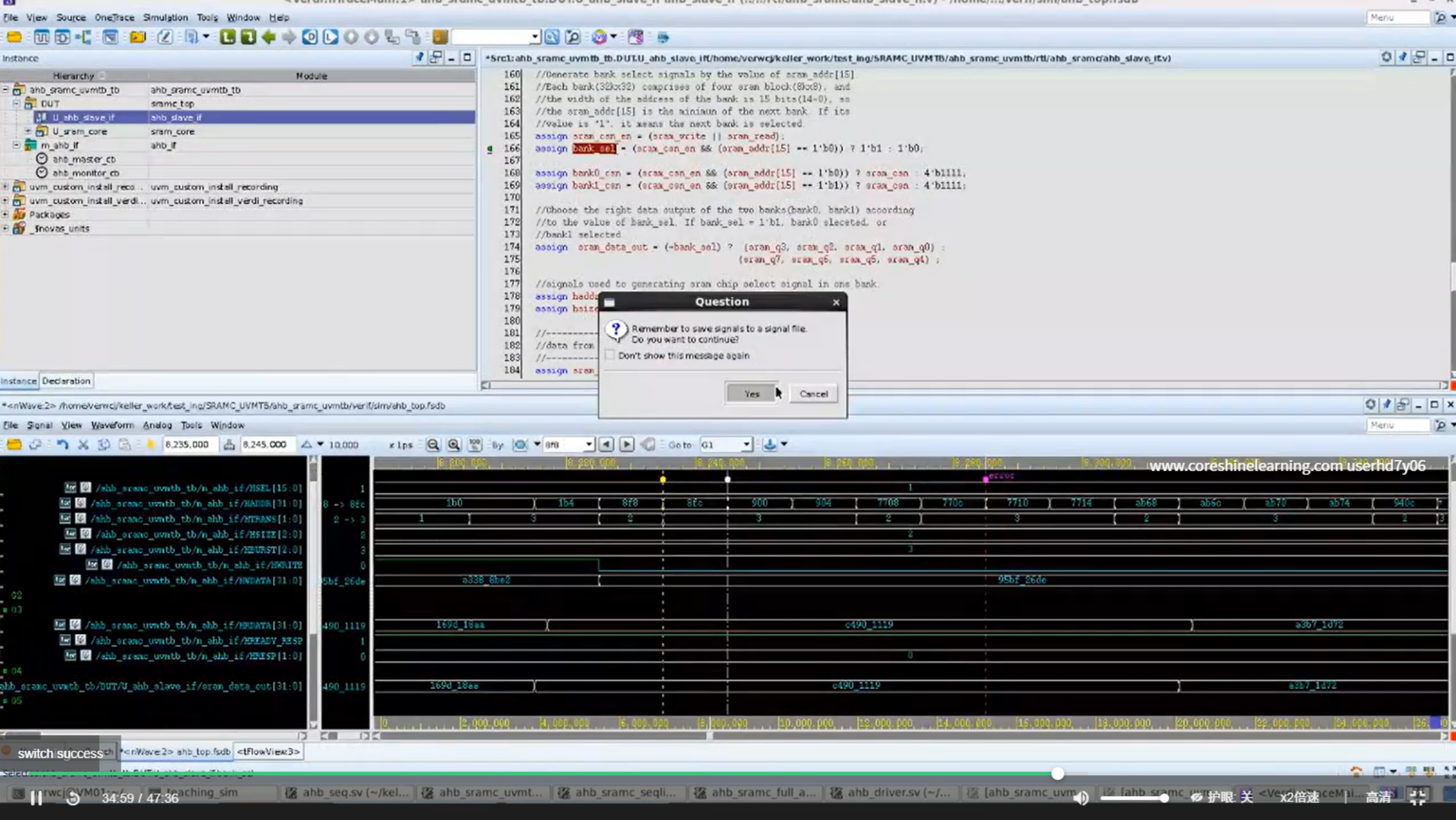
00, misma

www.coreshinelearning.com
userha7y06

switch success

166,58 67%

31:57 / 47:36 护眼: 关 x2倍速 高清



34:59 / 47:36
护眼:关
x2倍速
高清

```
[uvm_test_top.m_env.m_ahb_agent.m_monitor]  3833
[uvm_test_top.m_env.m_ahb_agent.m_sequencer.m_seq]     2
[uvm_test_top.m_env.m_scb]     1
$finish called from file "/qixin/eda/synopsys_6/vcsmx-2018.09/etc/uvm/base/uvm_root.svh", line 439.
$finish at simulation time              26415000
           V C S   S i m u l a t i o n   R e p o r t
Time: 26415000 ps
CPU Time:      2.690 seconds;       Data structure size:   0.6Mb
Sun Jul 30 16:41:20 2023
<verwcj-~/keller_work/test_ing/SRAMC_UVMTB/ahb_sramc_uvmtb/verif/sim> $
<verwcj-~/keller_work/test_ing/SRAMC_UVMTB/ahb_sramc_uvmtb/verif/sim> $
<verwcj-~/keller_work/test_ing/SRAMC_UVMTB/ahb_sramc_uvmtb/verif/sim> $make clean
rm -rf csrc simv* *.log DVE* *.vpd vc_hdrs.h *.key novas* verdiLog ahb_top.fsdb
<verwcj-~/keller_work/test_ing/SRAMC_UVMTB/ahb_sramc_uvmtb/verif/sim> $
<verwcj-~/keller_work/test_ing/SRAMC_UVMTB/ahb_sramc_uvmtb/verif/sim> $g ../../rtl/ahb_sramc/ahb_slave_if.v
<verwcj-~/keller_work/test_ing/SRAMC_UVMTB/ahb_sramc_uvmtb/verif/sim> $
<verwcj-~/keller_work/test_ing/SRAMC_UVMTB/ahb_sramc_uvmtb/verif/sim> $
<verwcj-~/keller_work/test_ing/SRAMC_UVMTB/ahb_sramc_uvmtb/verif/sim> $
<verwcj-~/keller_work/test_ing/SRAMC_UVMTB/ahb_sramc_uvmtb/verif/sim> $
<verwcj-~/keller_work/test_ing/SRAMC_UVMTB/ahb_sramc_uvmtb/verif/sim> $make all TEST=ahb_sramc_uvmtb_demo_test VERB=UVM_HIGH
vcs -sverilog -ntb_opts uvm -debug_access+all -timescale=1ns/1ns \
        -f ../../rtl/rtl.f \
  +incdir+../tb \
  +incdir+../agent/ahb \
  +incdir+../tests \
  +incdir+../tests/seqlib \
  +incdir+../env \
  ../agent/ahb/ahb_pkg.sv \
  ../agent/ahb/ahb_if.sv \
  ../env/ahb_sramc_uvmtb_env_pkg.sv \
  ../tests/ahb_sramc_uvmtb_test_pkg.sv \
  ../tb/ahb_sramc_uvmtb_tb.sv \
  -l comp.log -kdb
```

switch success

ahb_seq.sv (~/keller_work/test_ing/SRAMC_UVMTB/ahb_sramc_uvmtb/verif/agent/ahb) - GVIM3

File Edit Tools Syntax Buffers Window UVM Verilog Help

```
64
65    //repeat(10) begin
66    //  req = ahb_seq_item#(32,32)::type_id::create("req");
67    //  start_item(req);
68    //  if ( !req.randomize() with {req.trans == NONSEQ; req.burst == INCR4;} )
69    //    `uvm_error(get_type_name(), "Failed to randomize transaction")
70    //  finish_item(req);
71    //end
72
73    repeat(10) begin
74      req0 = ahb_seq_item#(32,32)::type_id::create("req0");
75      start_item(req0);
76      if ( !req0.randomize() with {req0.trans == NONSEQ; req0.burst == INCR4; req0.op == WRITE;} )
77        `uvm_error(get_type_name(), "Failed to randomize transaction")
78      finish_item(req0);
79
80      req1 = ahb_seq_item#(32,32)::type_id::create("req1");
81      start_item(req1);
82      if ( !req1.randomize() with {req1.trans == NONSEQ; req1.burst == INCR4; req1.op == READ; req1.haddr == req0.haddr; req1.size == req0.
    size;} )
83      //if ( !req1.randomize() with {req1.trans == NONSEQ; req1.burst == INCR4; req1.op == READ; req1.haddr == req0.haddr;} )
84        `uvm_error(get_type_name(), "Failed to randomize transaction")
85      finish_item(req1);
86    end
87    req = ahb_seq_item#(32,32)::type_id::create("req");
88    start_item(req);
89    if ( !req.randomize() with {req.trans == IDLE;} )
90      `uvm_error(get_type_name(), "Failed to randomize transaction")
91    finish_item(req);
92    `uvm_info(get_type_name(), "Default sequence completed", UVM_HIGH)
93  endtask : body
"../agent/ahb/ahb_seq.sv" 97L, 3002C written
```

www.coreshinelearning.com userhd7j06

switch success

83,6 94%

verwcj@VM01:... | teaching_sim | ahb_seq.sv (~/... | ahb_sramc_uv... | ahb_sramc_se... | ahb_sramc_full... | ahb_driver.sv (... | [ahb_sramc_uv... | <Verdi:nTrace... | ahb_sramc_uv... | ahb_sramc_uv...